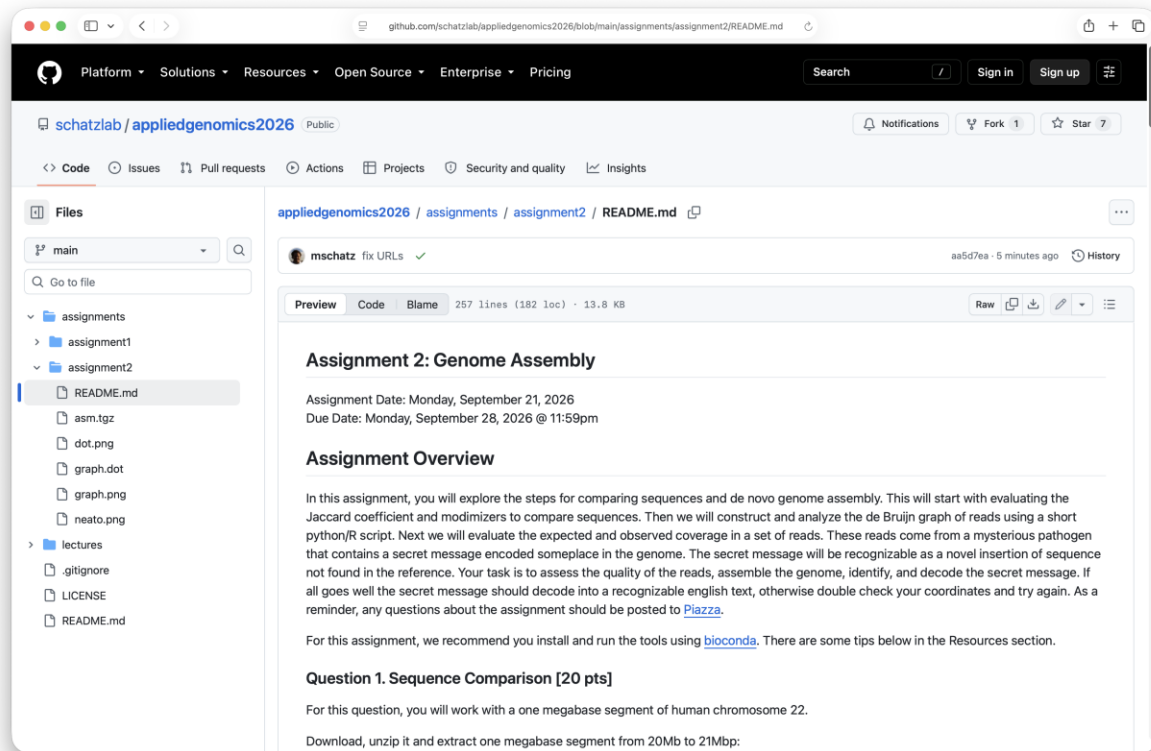


Introduction to Machine Learning

Nicole Brown & Michael Schatz
September 28, 2026

Assignment 2



The screenshot shows a GitHub web interface. The repository is 'schatzlab/appliedgenomics2026'. The file 'assignment2/README.md' is selected. The file content includes:

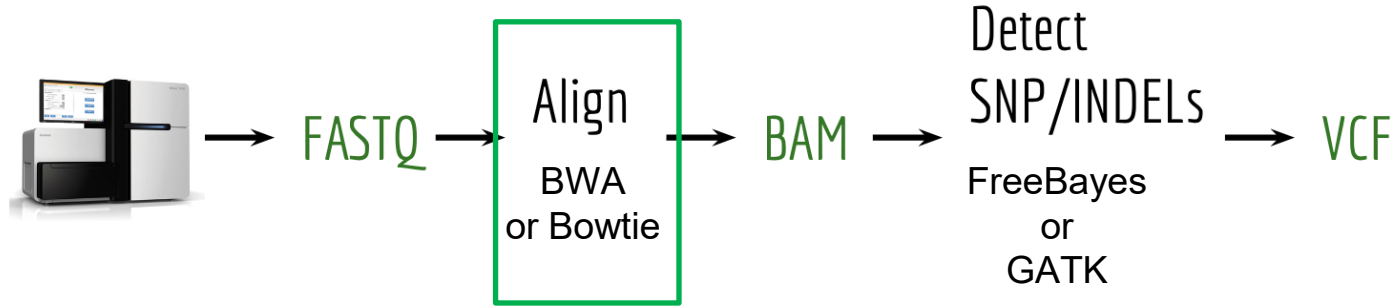
- Assignment 2: Genome Assembly**
- Assignment Date: Monday, September 21, 2026
- Due Date: Monday, September 28, 2026 @ 11:59pm
- Assignment Overview**
- A paragraph describing the assignment: "In this assignment, you will explore the steps for comparing sequences and de novo genome assembly. This will start with evaluating the Jaccard coefficient and modimizers to compare sequences. Then we will construct and analyze the de Bruijn graph of reads using a short python/R script. Next we will evaluate the expected and observed coverage in a set of reads. These reads come from a mysterious pathogen that contains a secret message encoded someplace in the genome. The secret message will be recognizable as a novel insertion of sequence not found in the reference. Your task is to assess the quality of the reads, assemble the genome, identify, and decode the secret message. If all goes well the secret message should decode into a recognizable english text, otherwise double check your coordinates and try again. As a reminder, any questions about the assignment should be posted to [Piazza](#)."
- A recommendation to install and run tools using [bioconda](#).
- Question 1. Sequence Comparison [20 pts]**
- Text: "For this question, you will work with a one megabase segment of human chromosome 22."
- Text: "Download, unzip it and extract one megabase segment from 20Mb to 21Mbp:"

<https://github.com/schatzlab/appliedgenomics2026/tree/main/assignments/assignment2>
Due end of day on Monday Sept 28 (right before midnight)

Variant Calling Overview



Variant Calling Overview



Finding Alignments: Brute Force



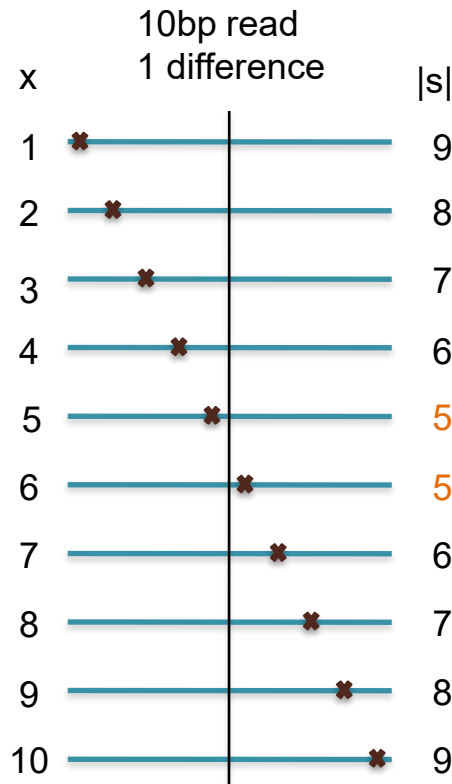
- Brute Force:
 - At every possible offset in the genome:
 - Do all of the characters of the query match?
- To align the query GATTACA to a 3 Gbp reference:
 - Genome length = n [3B]
 - Query length = m [7]
 - Comparisons: $(n-m+1) * m$ [21B]
- Overall runtime: $O(nm)$
 - [How long would it take if we double the genome size, read length?]
 - [How long would it take if we double both?]

Finding Alignments: Seed-and-Extend

Theorem: An alignment of a sequence of length m with at most k differences **must** contain an exact match at least $s = m/(k+1)$ bp long

(Baeza-Yates and Perleberg, 1996)

- Proof: Pigeonhole principle
 - 1 pigeon can't fill 2 holes
- Seed-and-extend search
 - Use an index to rapidly find short exact alignments to seed longer in-exact alignments
 - BLAST, MUMmer, Bowtie, BWA, SOAP, ...
 - Specificity depends on seed length
 - Guaranteed sensitivity for k differences
 - Also finds some (but not all) lower quality alignments <- heuristic



Finding Alignments: Suffix Array

- Strategy 2: Binary search
 - Compare to the middle, refine as higher or lower
- Searching for GATTACA
 - $Lo = 1; Hi = 15; Mid = (1+15)/2 = 8$
 - $Middle = Suffix[8] = CC$
=> Higher: $Lo = Mid + 1$
 - $Lo = 9; Hi = 15; Mid = (9+15)/2 = 12$
 - $Middle = Suffix[12] = TACC$
=> Lower: $Hi = Mid - 1$
 - $Lo = 9; Hi = 11; Mid = (9+11)/2 = 10$
 - $Middle = Suffix[10] = GATTACC$
=> Lower: $Hi = Mid - 1$
 - $Lo = 9; Hi = 9; Mid = (9+9)/2 = 9$
 - $Middle = Suffix[9] = GATTACA...$
=> Match at position 2!

#	Sequence	Pos
1	ACAGATTACC...	6
2	ACC...	13
3	AGATTACC...	8
4	ATTACAGATTACC...	3
5	ATTACC...	10
6	C...	15
7	CAGATTACC...	7
8	CC...	14
9	GATTACAGATTACC...	2
10	GATTACC...	9
11	TACAGATTACC...	5
12	TACC...	12
13	TGATTACAGATTACC...	1
14	TTACAGATTACC...	4
15	TTACC...	11

Lo
Hi
→

Alignment Similarity Metrics

- Hamming distance

- Count the number of substitutions to transform one string into another

MIKESCHATZ

| | X | | XXXX |

MICESHATZZ

5

- Edit distance

- The minimum number of substitutions, insertions, or deletions to transform one string into another

MIKESCHAT-Z

| | X | | X | | | X |

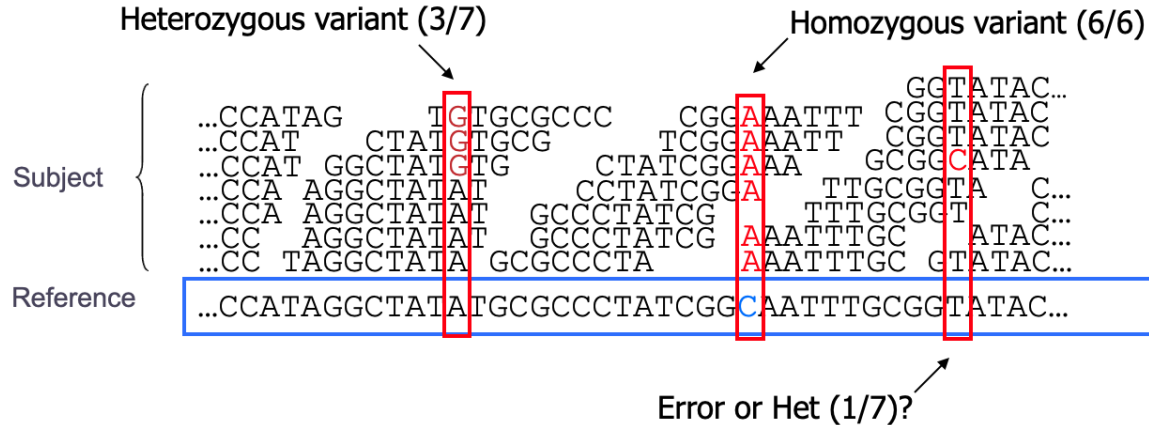
MICES-HATZZ

3

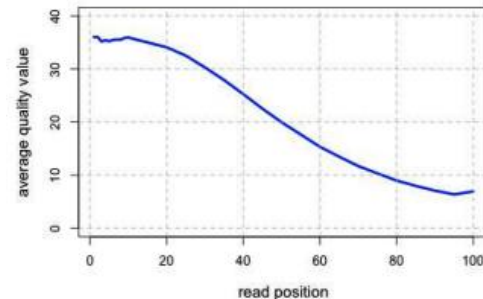
Variant Calling Overview



Variant Calling: Genotyping Theory



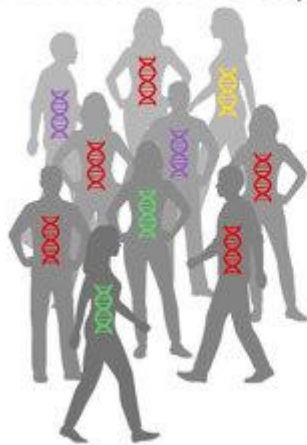
- If there were no sequencing errors, identifying SNPs would be very easy: any time a read disagrees with the reference, it must be a variant!
- Sequencing instruments make mistakes
 - Quality of read decreases over the read length
- A single read differing from the reference is probably just an error, but it becomes more likely to be real as we see it multiple times



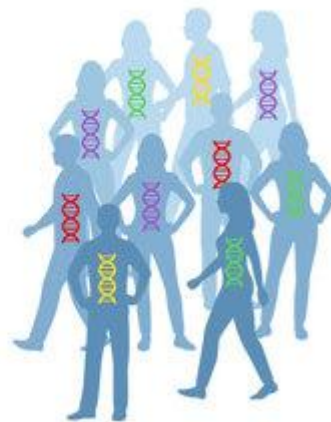
Variant Calling: Data Analysis

- We want to find methods to:
 - **Cluster** samples with similar variants together
 - **Predict** phenotypes based on variants

Non-Disease Group



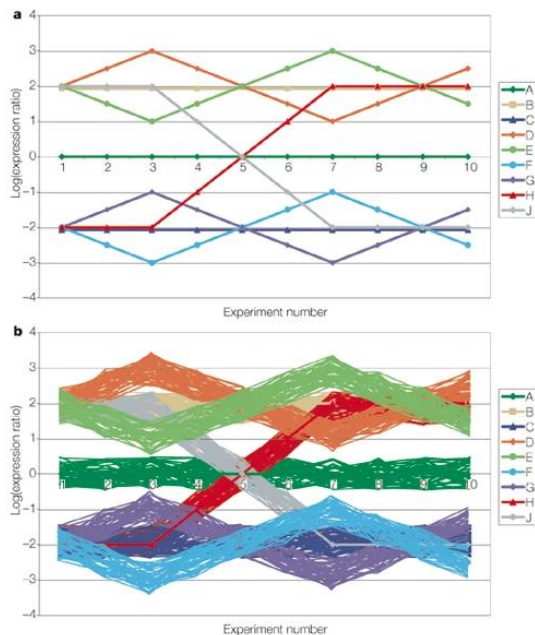
Disease Group



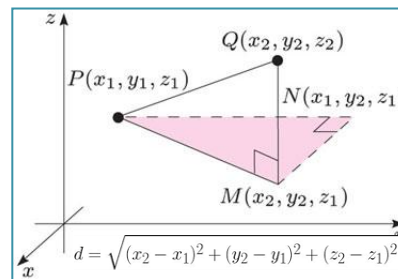
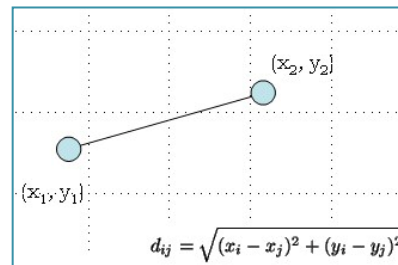
Clustering & Dimensionality Reduction

Clustering Refresher

Euclidean Distance



Nature Reviews | Genetics

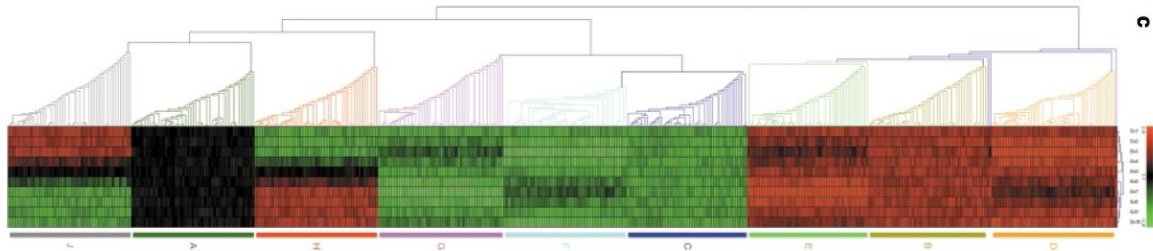
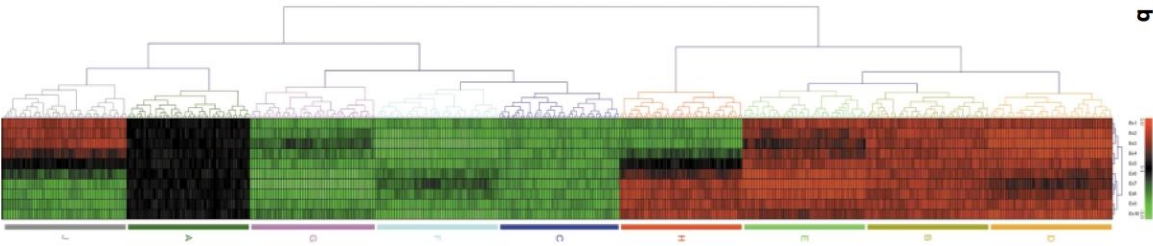
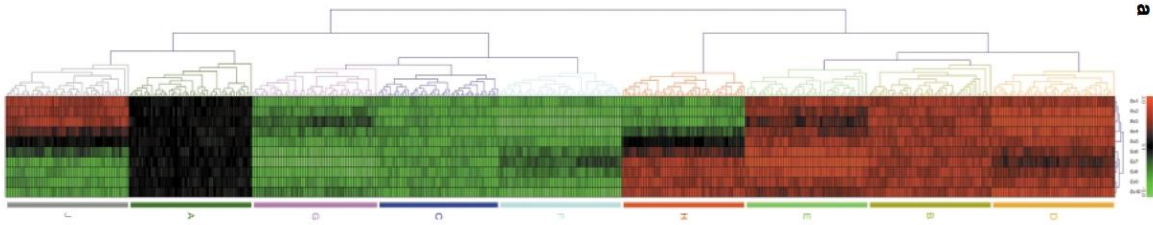


$$d(p, q) = d(q, p) = \sqrt{(q_1 - p_1)^2 + (q_2 - p_2)^2 + \dots + (q_n - p_n)^2} = \sqrt{\sum_{i=1}^n (q_i - p_i)^2}.$$

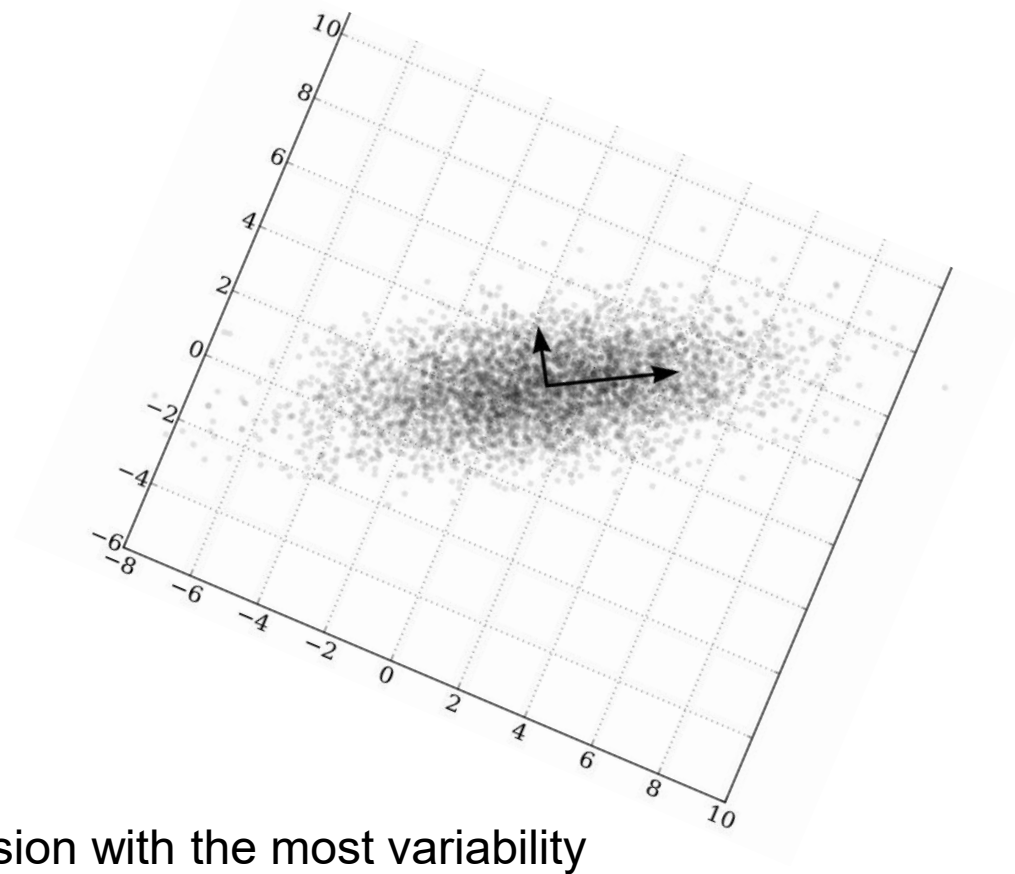
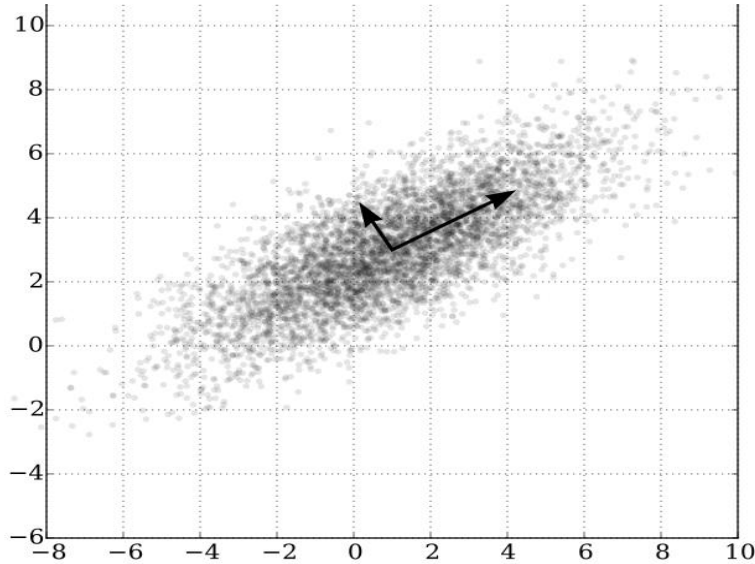
Computational genetics: Computational analysis of microarray data

Quackenbush (2001) *Nature Reviews Genetics*. doi:10.1038/35076576

Hierarchical Clustering



Principal Component Analysis (PCA)



PC1: “New X”- The dimension with the most variability
PC2: “New Y”- The dimension with the second most variability

Principal Components Analysis (PCA)

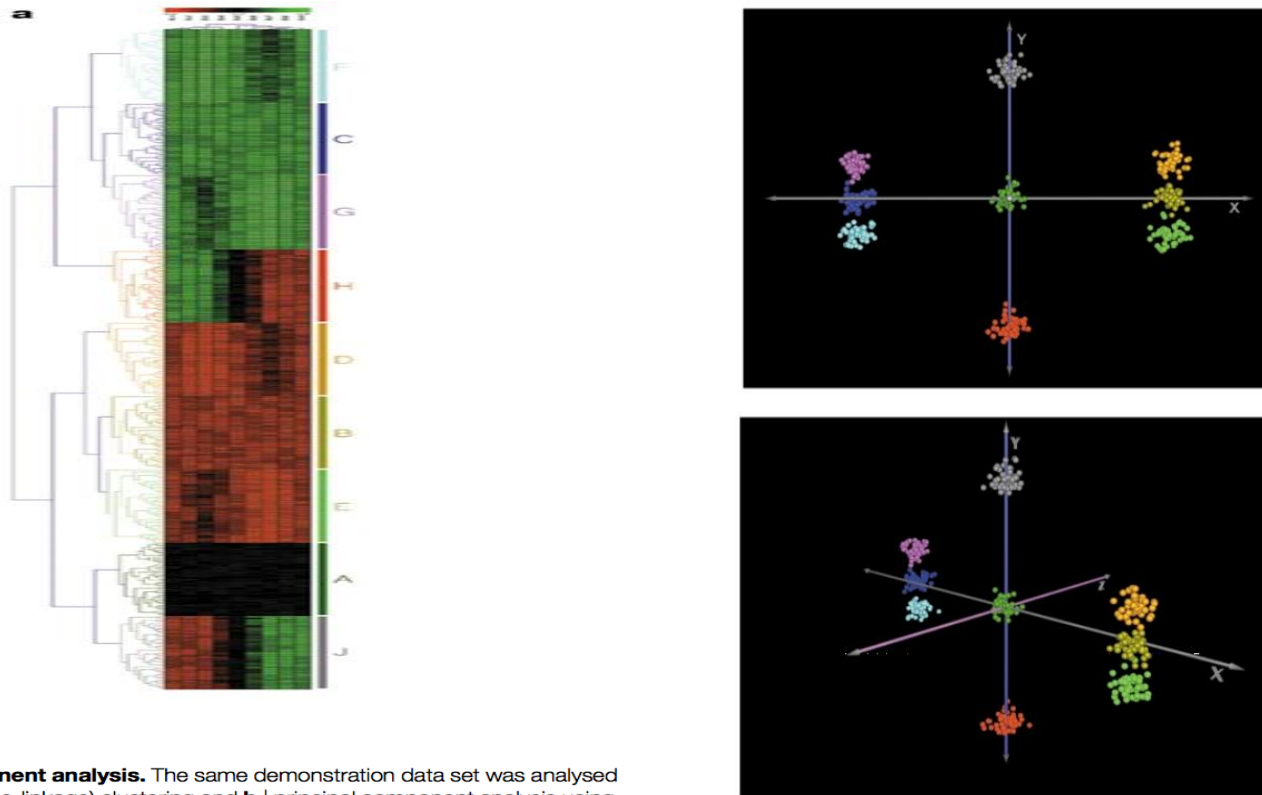


Figure 4 | **Principal component analysis.** The same demonstration data set was analysed using **a** | hierarchical (average-linkage) clustering and **b** | principal component analysis using Euclidean distance, to show how each treats the data, with genes colour coded on the basis of hierarchical clustering results for comparison.

MNIST



- 70,000 images, each labeled with the correct digit (0-9)
 - 28 x 28 thumbnail images of handwritten digits
 - Grayscale image with 256 possible values (784 bytes per image)
- Widely used dataset for image classification
 - Very simple, the categories are very easy to understand

MNIST: PCA



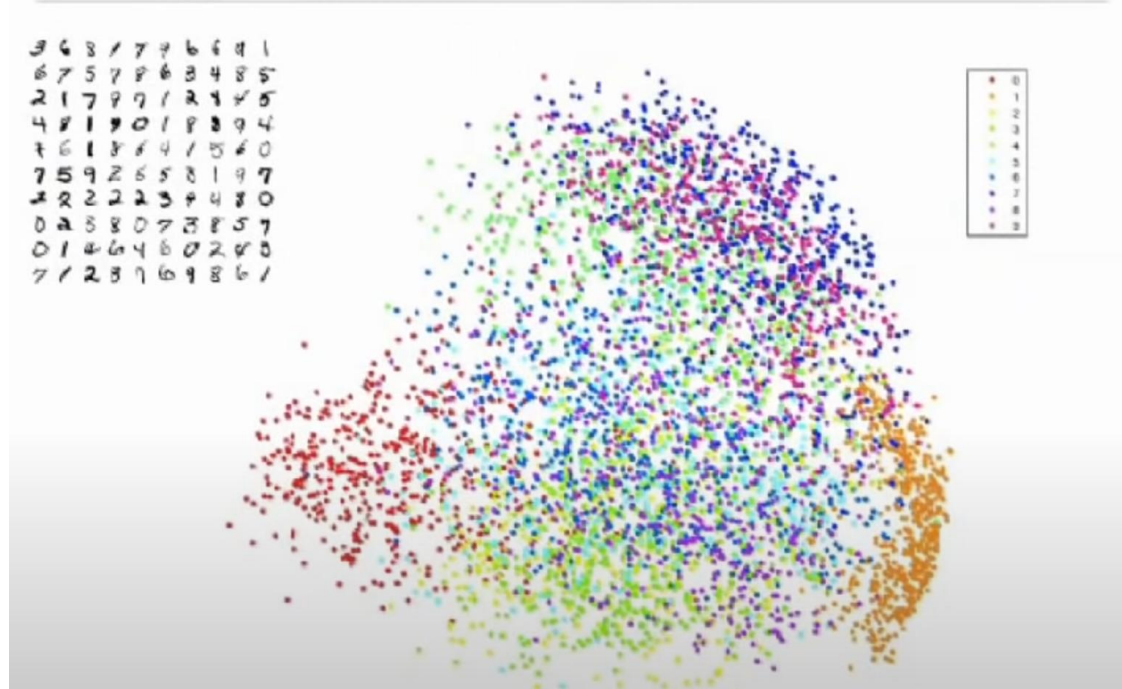
Principal Components Analysis

Each 28x28 image can be considered a point in 784 dimensional space!

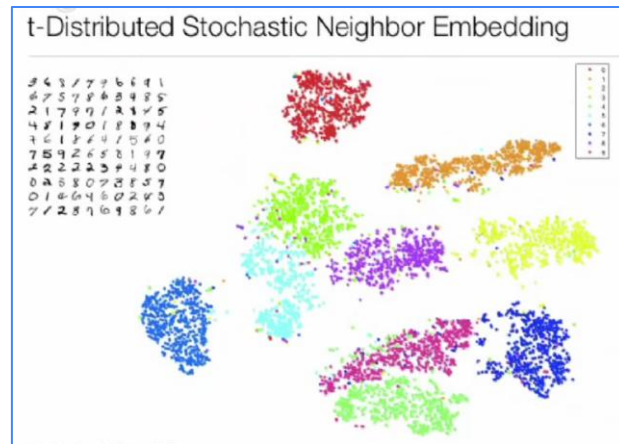
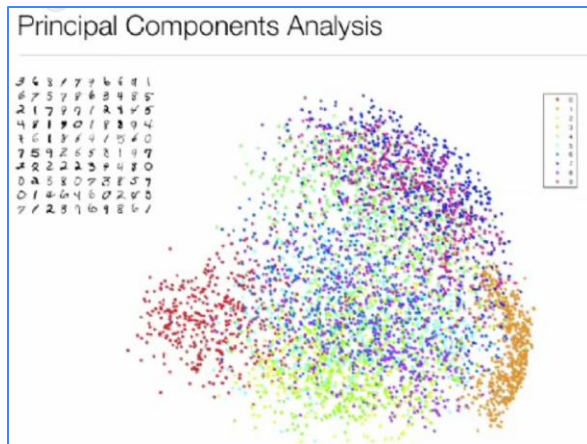
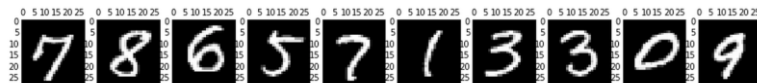
PCA roughly groups digits together

- 0 (red) in bottom left
- 1 (orange) in bottom right

But very incomplete separation, lots of mixing of colors



MNIST: PCA and t-SNE



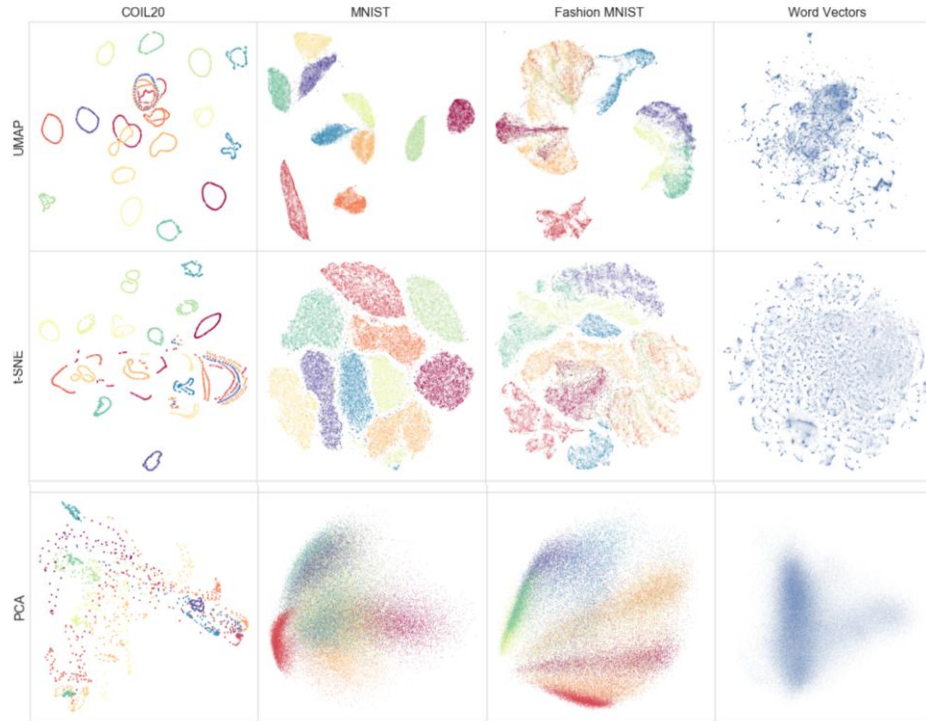
t-distributed Stochastic Neighborhood Embedding

- Non-linear dimensionality reduction technique: distances are only locally meaningful
- Rather than Euclidean distances, for each point fits a Gaussian kernel to fit the nearest N neighbors (perplexity) that define the probabilities that two points should be close together
- Using an iterative spring embedding system to place high probability points nearby

Visualizing Data Using t-SNE

<https://www.youtube.com/watch?v=RJVL80Gg3IA>

UMAP



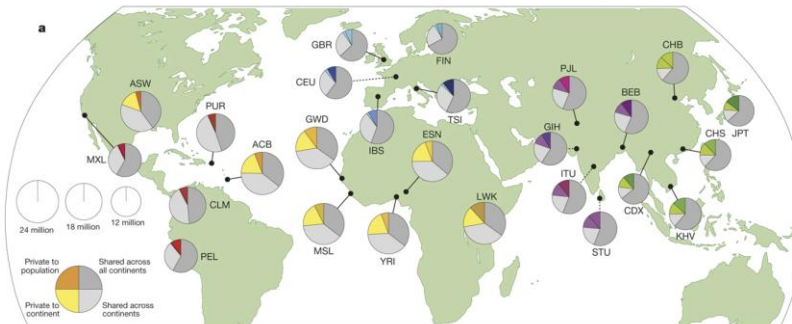
UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction

McInnes et al (2018) arXiv. 1802.03426

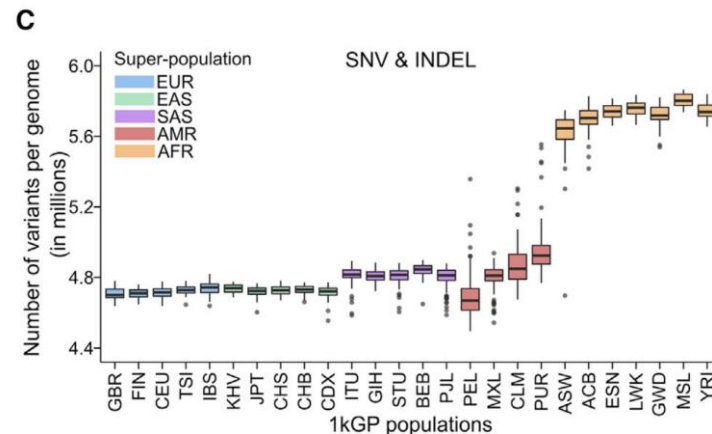
<https://www.youtube.com/watch?v=nq6iPZVUxZU>

<https://towardsdatascience.com/how-exactly-umap-works-13e3040e1668>

1000 Genomes Project

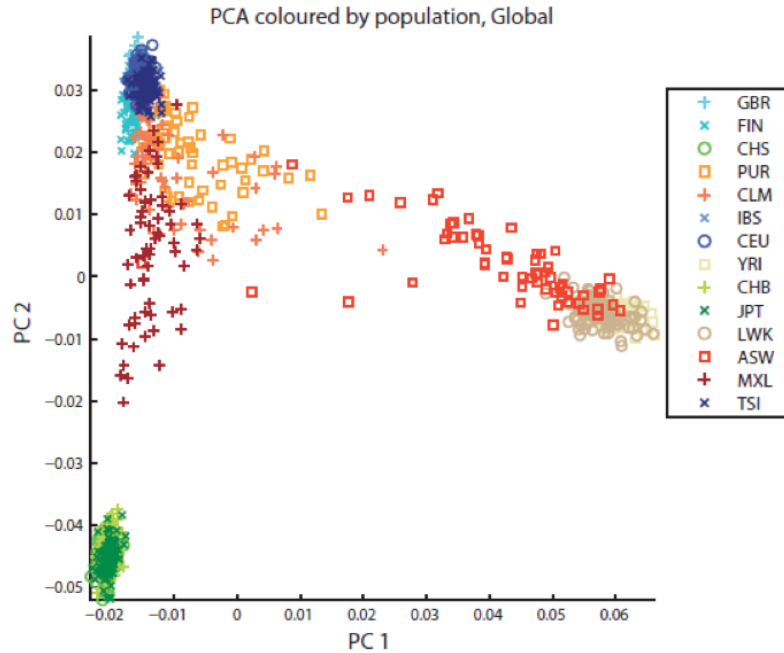


EAS	CHB	Han Chinese	Han Chinese in Beijing, China
	JPT	Japanese	Japanese in Tokyo, Japan
	CHS	Southern Han Chinese	Han Chinese South
	CDX	Dai Chinese	Chinese Dai in Xishuangbanna, China
	KHV	Kinh Vietnamese	Kinh in Ho Chi Minh City, Vietnam
	CHD	Denver Chinese	Chinese in Denver, Colorado (pilot 3 only)
EUR	CEU	CEPH	Utah residents (CEPH) with Northern and Western European ancestry
	TSI	Tuscan	Toscani in Italy
	GBR	British	British in England and Scotland
	FIN	Finnish	Finnish in Finland
	IBS	Spanish	Iberian populations in Spain
AFR	YRI	Yoruba	Yoruba in Ibadan, Nigeria
	LWK	Luhya	Luhya in Webuye, Kenya
	GWD	Gambian	Gambian in Western Division, The Gambia
	MSL	Mende	Mende in Sierra Leone
	ESN	Esan	Esan in Nigeria
AMR	ASW	African-American SW	African Ancestry in Southwest US
	ACB	African-Caribbean	African Caribbean in Barbados
	MXL	Mexican-American	Mexican Ancestry in Los Angeles, California
	PUR	Puerto Rican	Puerto Rican in Puerto Rico
	CLM	Colombian	Colombian in Medellin, Colombia
	PEL	Peruvian	Peruvian in Lima, Peru
SAS	GIH	Gujarati	Gujarati Indian in Houston, TX
	PJL	Punjabi	Punjabi in Lahore, Pakistan
	BEB	Bengali	Bengali in Bangladesh
	STU	Sri Lankan	Sri Lankan Tamil in the UK
	ITU	Indian	Indian Telugu in the UK



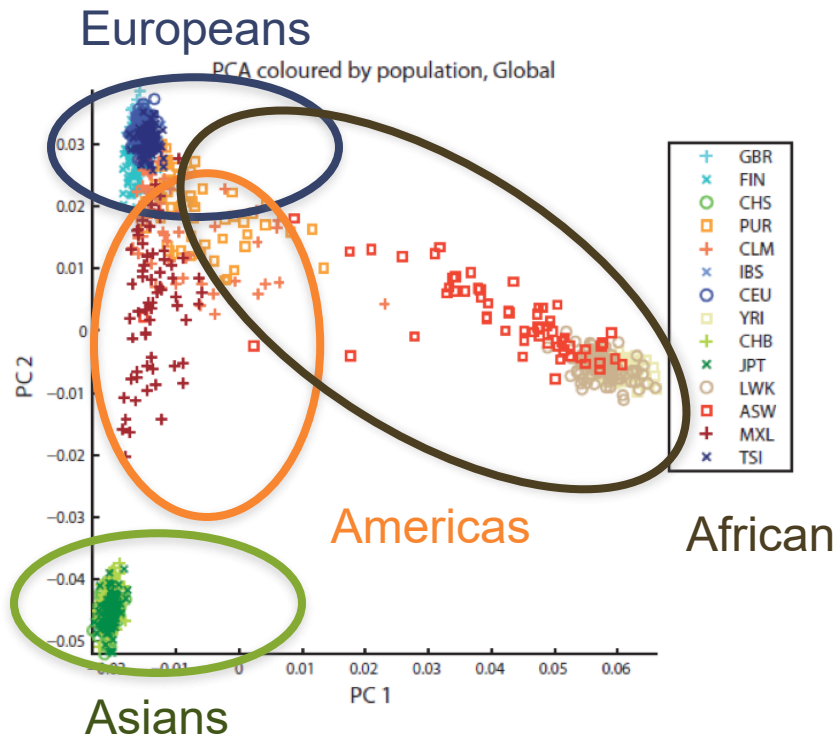
High-coverage whole-genome sequencing of the expanded 1000 Genomes Project cohort including 602 trios
 Byrska-Bishop et al. (2022) Cell. doi: 10.1016/j.cell.2022.08.004

Variation across populations



	Person1	Person2	Person3	...
SNP1	0/0	0/1	0/1	
SNP2	0/1	0/1	0/1	
SNP3	0/0	0/0	1/1	
SNP4	1/1	0/1	1/1	
SNP5	0/0	1/1	1/1	
SNP5	0/0	0/0	0/1	
...				

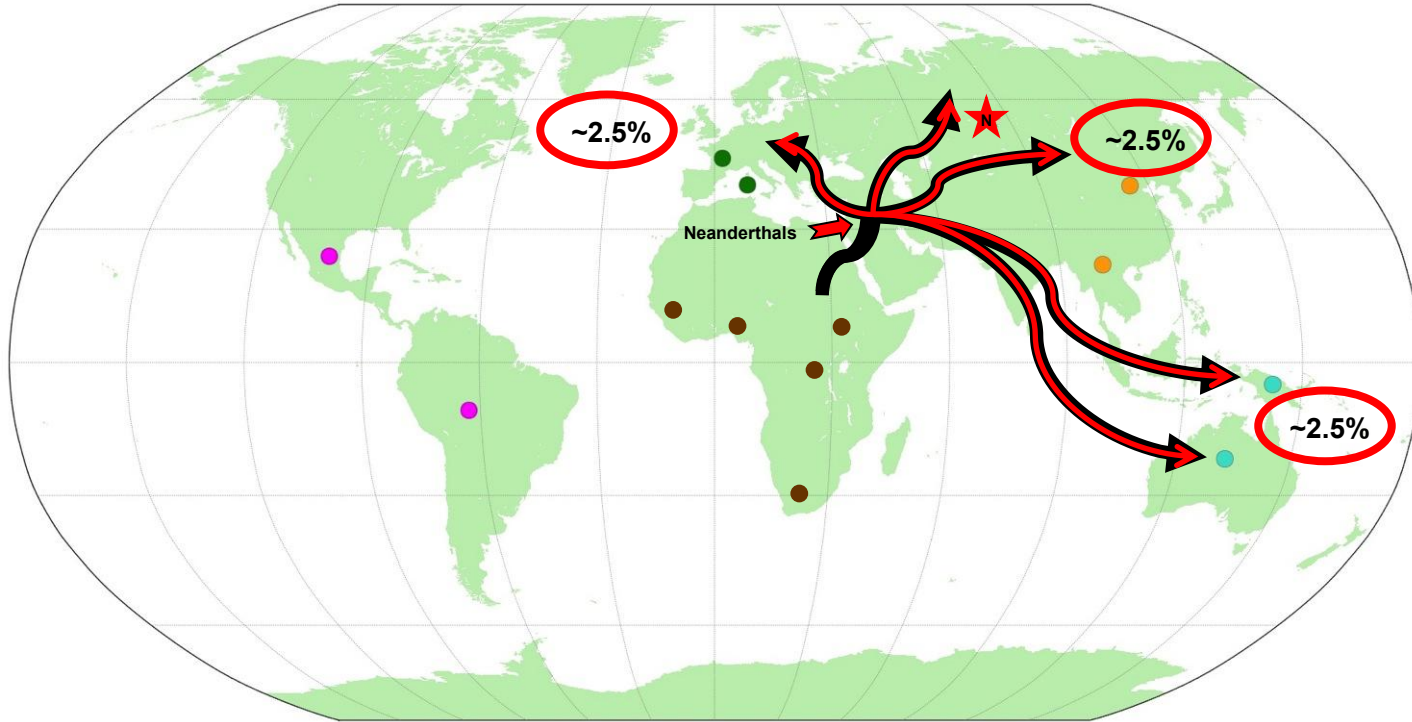
Variation across populations



	Person1	Person2	Person3	...
SNP1	0/0	0/1	0/1	
SNP2	0/1	0/1	0/1	
SNP3	0/0	0/0	1/1	
SNP4	1/1	0/1	1/1	
SNP5	0/0	1/1	1/1	
SNP5	0/0	0/0	0/1	
...				

PC1 is largely correlated with African ancestry:
Human genetic diversity is dominated by African ancestry

Human Migrations



As modern humans migrated out of Africa, they apparently interbred with Neanderthals so we see their alleles across the rest of the world and carry about 2.5% of their genome with us!



1,686,213 plays | Svante Pääbo | TEDGlobal 2011 • July 2011

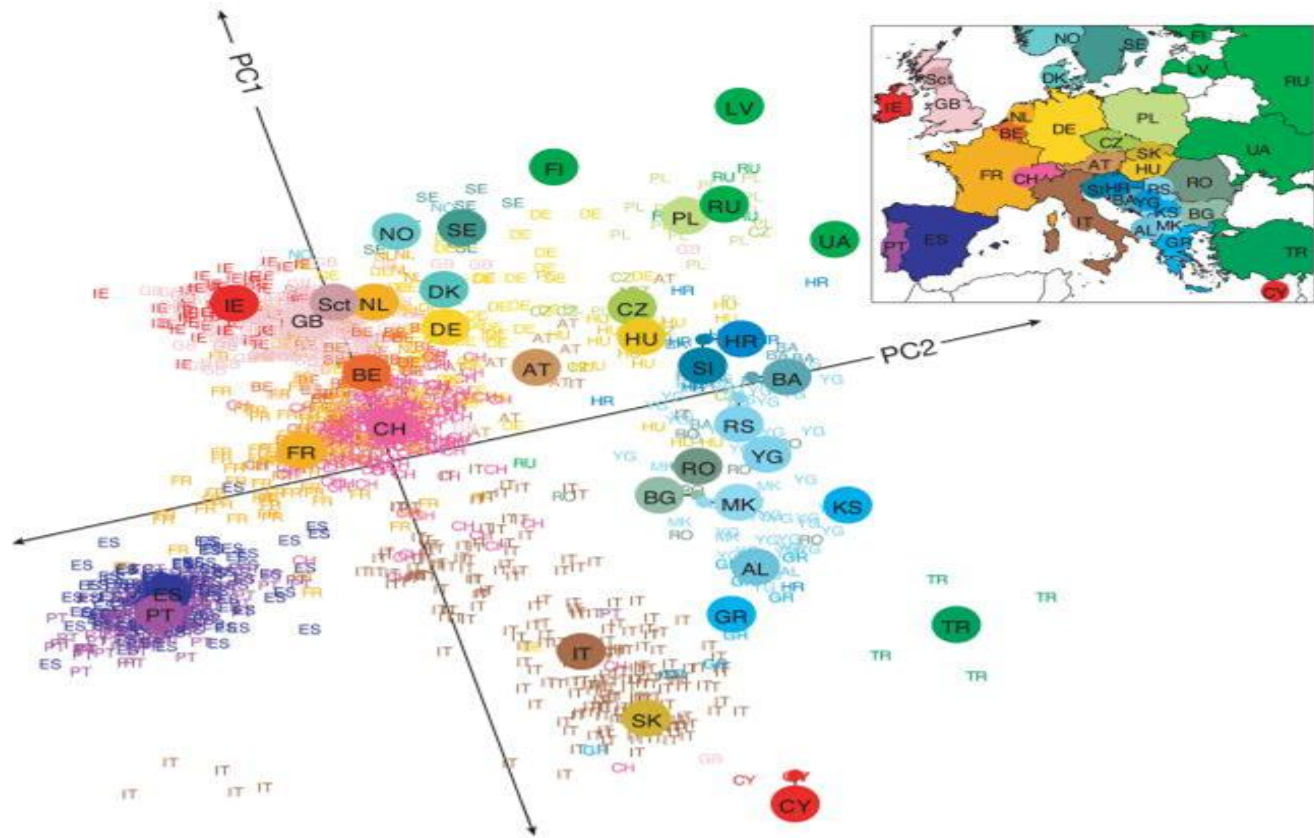
[Read transcript](#)

Science, DNA, Evolution, Biology.

May 2013
**Bring back the
woolly mammoth!**
Hendrik Poinar

Science, DNA, Evolution, Biology

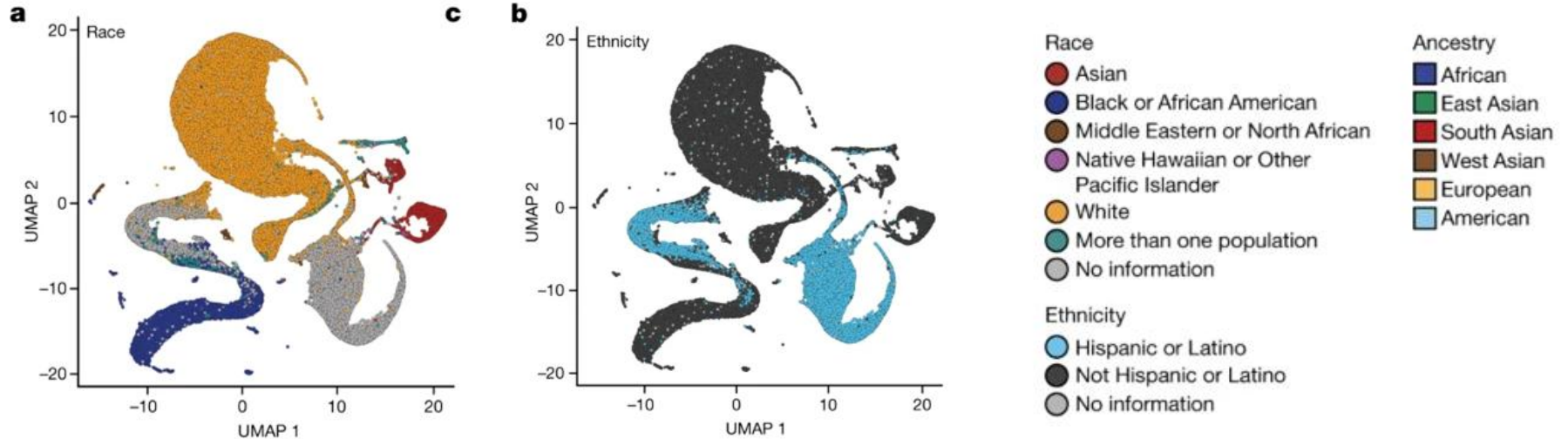
a



Genes mirror geography within Europe

Novembre et al (2008) Nature. doi: 10.1038/nature07331

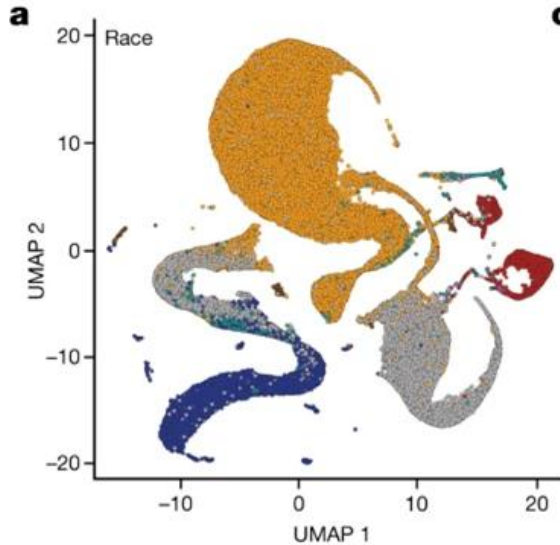
UMAP of Human Variation



Genomic data in the All of Us Research Program

The All of Us Research Program Genomics Investigators (2024) Nature. doi.org/10.1038/s41586-023-06957-x

~~UMAP of Human Variation~~



c “It's a pity that All of Us used UMAP to visualize ancestry variation in their new marker paper, out today in Nature. The UMAP algorithm, by design, exaggerates the distinctiveness of the most frequent ancestries, a message that can be misinterpreted by the public. UMAP pulls unusual genotypes towards the majority clusters; in particular it fails to represent admixture in a sensible way (admixture is fundamentally additive, while UMAP is not). In this setting the messiness of Admixture or PCA plots yield a better reflection of the data”

Jonathan Pritchard
Stanford

Genomic data in the All of Us Research Program

The All of Us Research Program Genomics Investigators (2024) Nature. doi.org/10.1038/s41586-023-06957-x

Introduction to Machine Learning: Classification & Prediction

Classification Problems

Supervised learning problem:

- Given (many) labeled training examples, develop a classifier that will accurately predict an output
- Each example contains **features** used to form a prediction

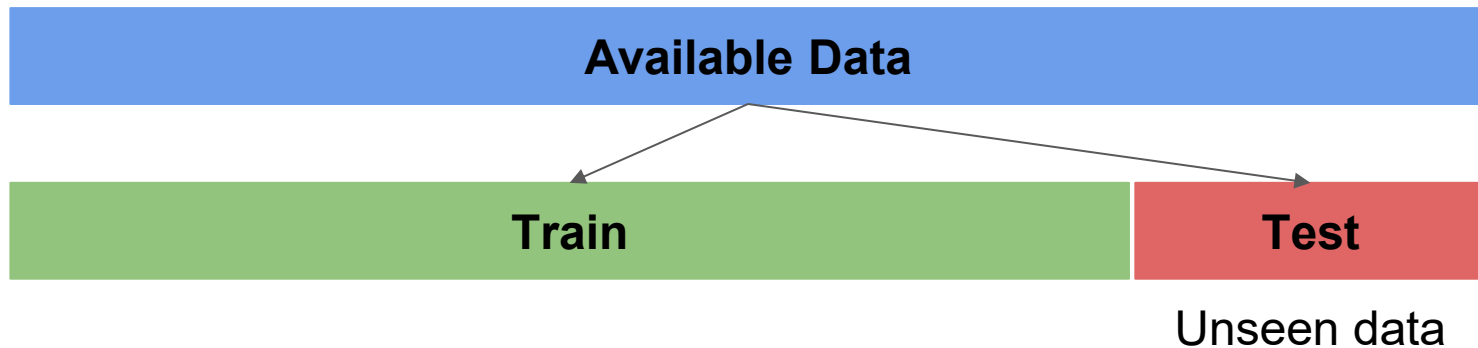
Examples:

- Given a pileup of reads, predict if there really is a variant
- Given SNVs in a person, predict ancestry (or relatedness)
- Given SNVs/expression, predict disease risk
- Given SNVs/expression & disease status, predict treatment
- Given an image (photograph, xray, MRI), predict what it is (object, tumor, etc)
- Given English text, classify the sentiment (happy/sad) or predict the next word
- ...

Training datasets and cross validation

We need to do two things with the data:

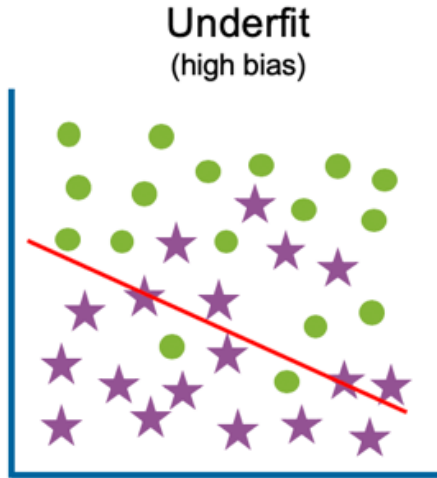
1. Estimate the parameters for the ML method (**Training**)
2. Evaluate how well the ML method works (**Testing**)



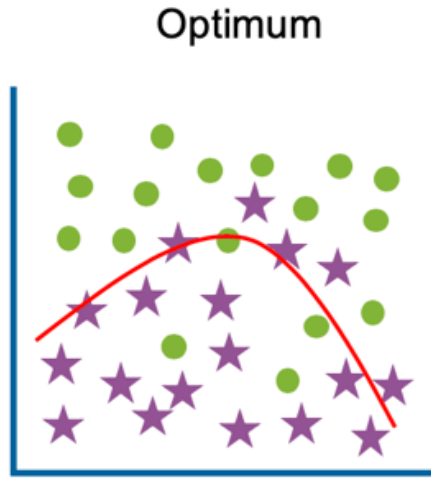
But how do we optimize the model?

Overfitting

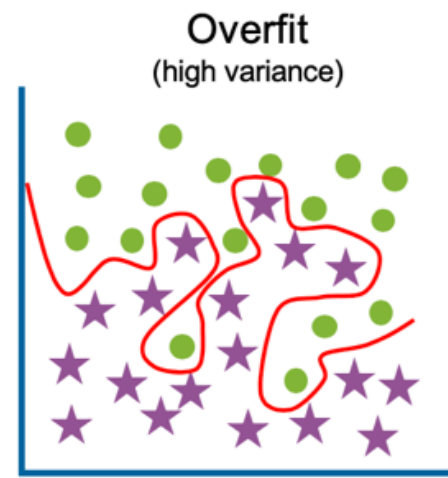
Model is very accurate on training data but can't generalize to new data



High training error
High test error



Low training error
Low test error

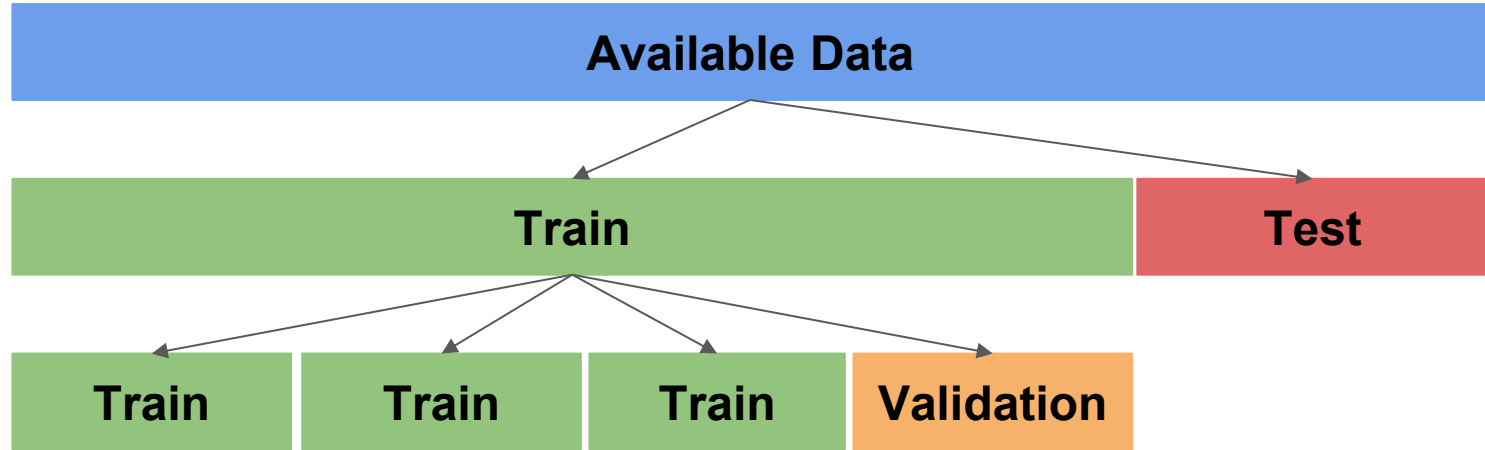


Low training error
High test error

Cross Validation

We can split our training data into 75% train and 25% validation

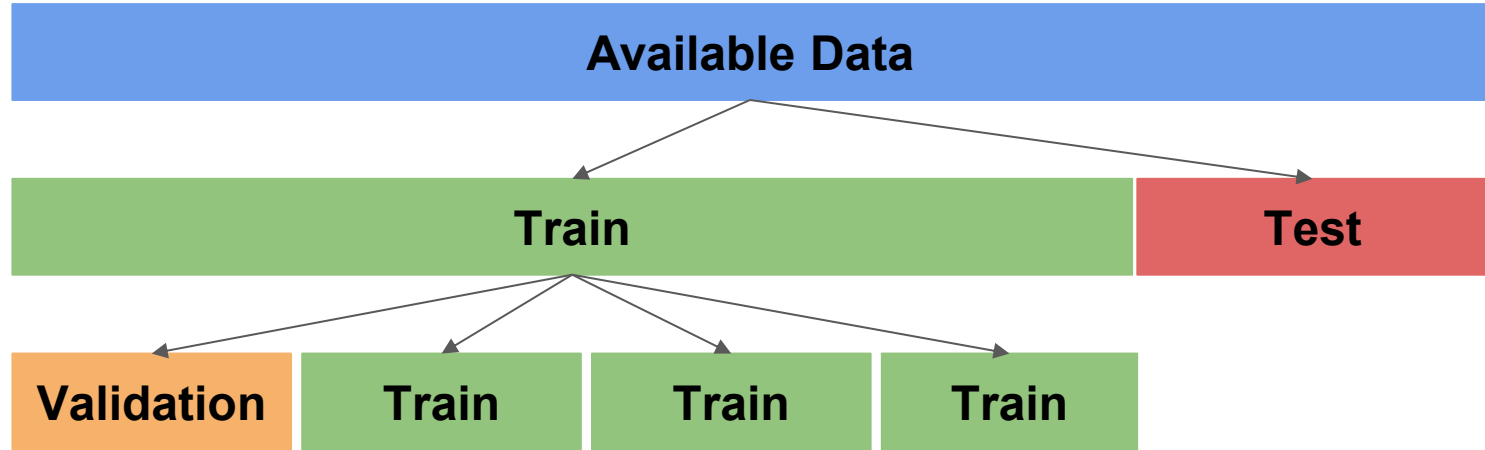
- But which block do we use for validation?



Cross Validation

We can split our training data into 75% train and 25% validation

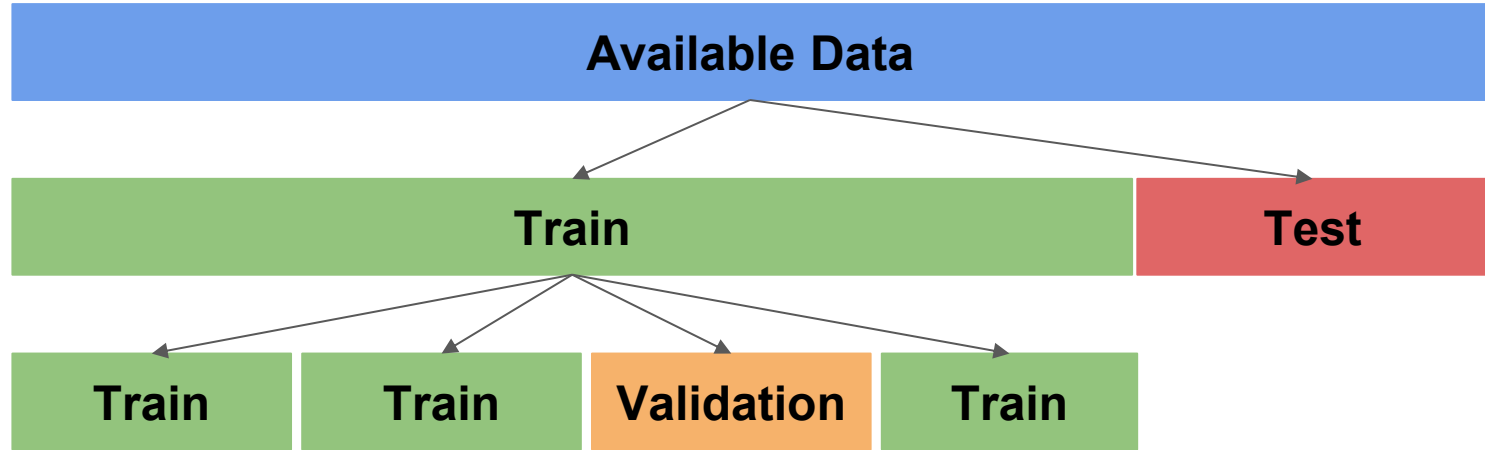
- But which block do we use for validation?
 - Why not all of them?



Cross Validation

We can split our training data into 75% train and 25% validation

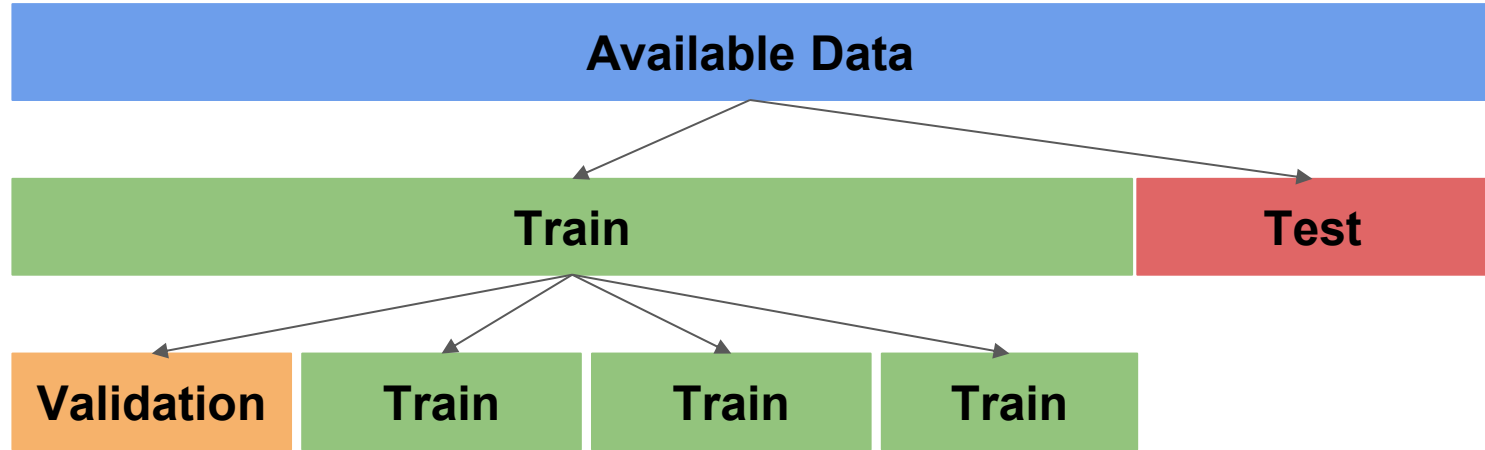
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Cross Validation

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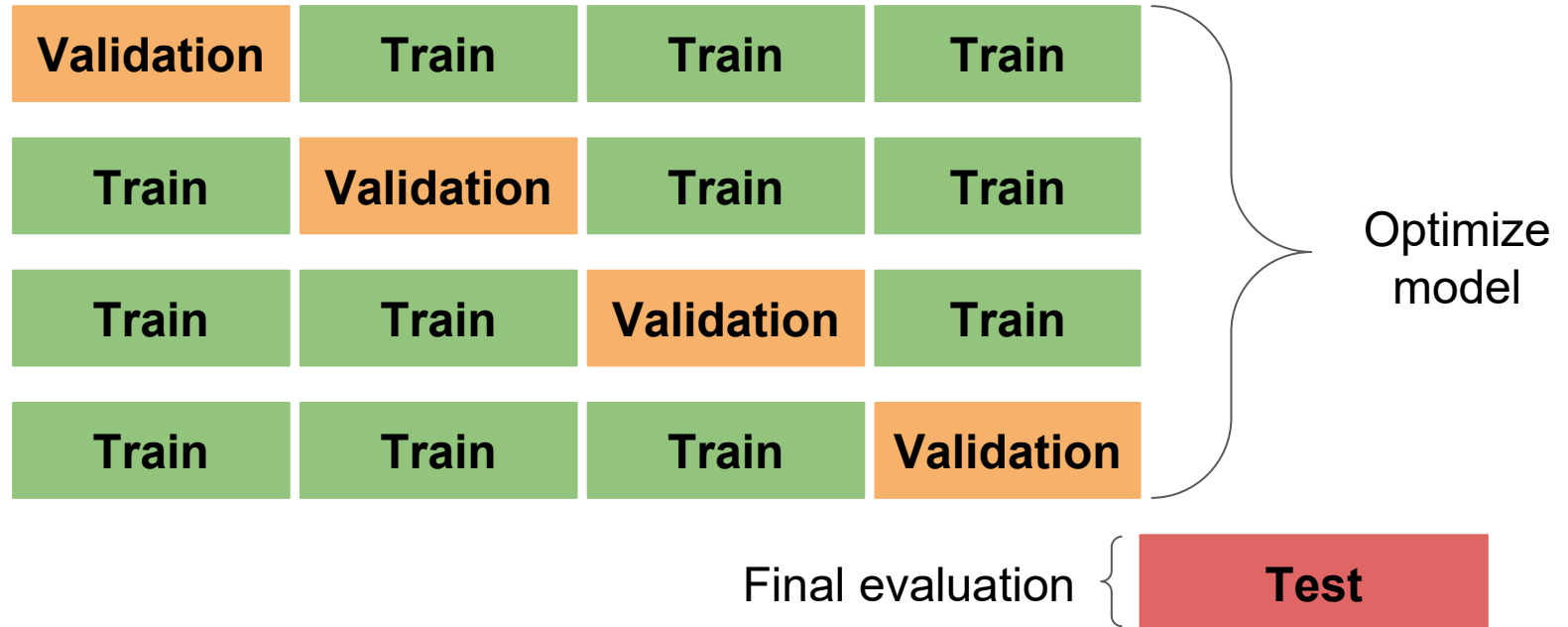
- But which block do we use for validation?
 - Why not all of them?



Cross Validation

k-Fold cross validation

- Split training data into k splits



Evaluation Metrics

Confusion matrix:

		True Class	
		Positive	Negative
Predicated Class	Positive	TP	FP
	Negative	FN	TN

Evaluation Metrics

Confusion matrix:

		True Class	
		Positive	Negative
Predicated Class	Positive	TP	FP
	Negative	FN	TN

Accuracy:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Evaluation Metrics

Confusion matrix:

		True Class	
		Positive	Negative
Predicated Class	Positive	TP	FP
	Negative	FN	TN

Accuracy:

- But what if the classes are unbalanced?

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Evaluation Metrics

		True Class	
		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

Precision:

- Of the things the model predicted as positive, how many were actually positive?
Use when false positives are of higher concern

$$Precision = \frac{TruePositive}{TruePositive + FalsePositive}$$

Evaluation Metrics

		True Class	
		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

Precision:

- Of the things the model predicted as positive, how many were actually positive?
Use when false positives are of higher concern

$$Precision = \frac{TruePositive}{TruePositive + FalsePositive}$$

Recall (sensitivity):

- Of all the real positives, how many did the model find?
Use when false negatives are of higher concern

$$Recall = \frac{TruePositive}{TruePositive + FalseNegative}$$

Evaluation Metrics

		True Class	
		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

Precision:

- Of the things the model predicted as positive, how many were actually positive? Use when false positives are of higher concern

$$\text{Precision} = \frac{\text{TruePositive}}{\text{TruePositive} + \text{FalsePositive}}$$

Recall (sensitivity):

- Of all the real positives, how many did the model find?
Use when false negatives are of higher concern

$$\text{Recall} = \frac{\text{TruePositive}}{\text{TruePositive} + \text{FalseNegative}}$$

F1 Score:

- Combines precision/recall (harmonic mean)

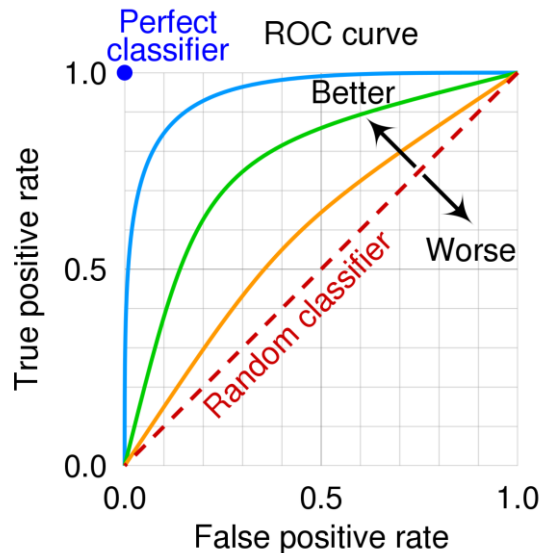
$$F1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Evaluation Metrics

		True Class	
		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

AUC-ROC

- Receiver Operator Characteristic (ROC) plots the true positive rate against the false positive rate
- Area Under the Curve (AUC) measure how the model distinguishes classes



Scikit-learn



Install User Guide API Examples Community [More](#)



1.9.1 (stable)

scikit-learn

Machine Learning in Python

Getting Started

Release Highlights for 1.9

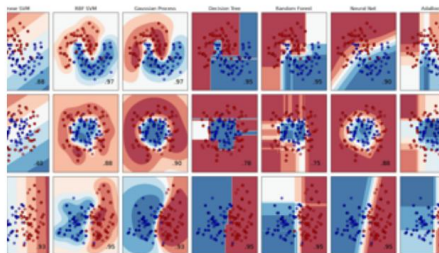
- Simple and efficient tools for predictive data analysis
- Accessible to everybody, and reusable in various contexts
- Built on NumPy, SciPy, and matplotlib
- Open source, commercially usable - BSD license

Classification

Identifying which category an object belongs to.

Applications: Spam detection, image recognition.

Algorithms: [Gradient boosting](#), [nearest neighbors](#), [random forest](#), [logistic regression](#), and [more...](#)



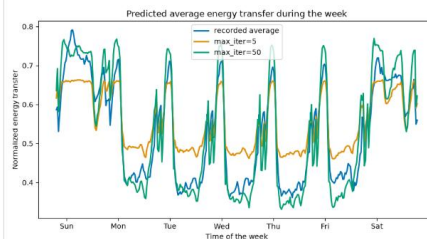
Examples

Regression

Predicting a continuous-valued attribute associated with an object.

Applications: Drug response, stock prices.

Algorithms: [Gradient boosting](#), [nearest neighbors](#), [random forest](#), [ridge](#), and [more...](#)



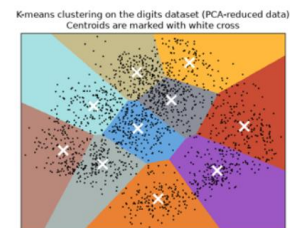
Examples

Clustering

Automatic grouping of similar objects into sets.

Applications: Customer segmentation, grouping experiment outcomes.

Algorithms: [k-Means](#), [HDBSCAN](#), [hierarchical clustering](#), and [more...](#)



Examples

Dimensionality reduction

Reducing the number of random variables to consider.

Model selection

Comparing, validating and choosing parameters and

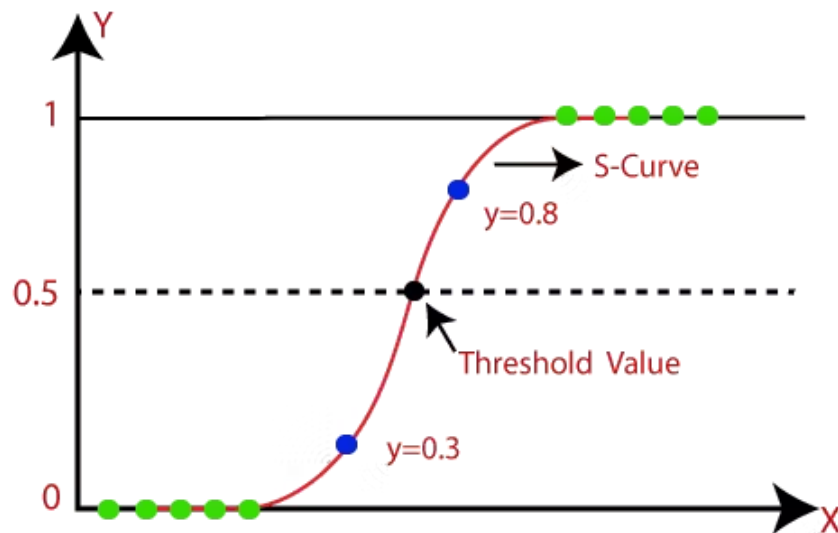
Preprocessing

Feature extraction and normalization.

Logistic Regression

Logistic regression assigns a weight to each feature to form a prediction

$$\hat{y}(w, x) = w_0 + w_1x_1 + \dots + w_px_p$$

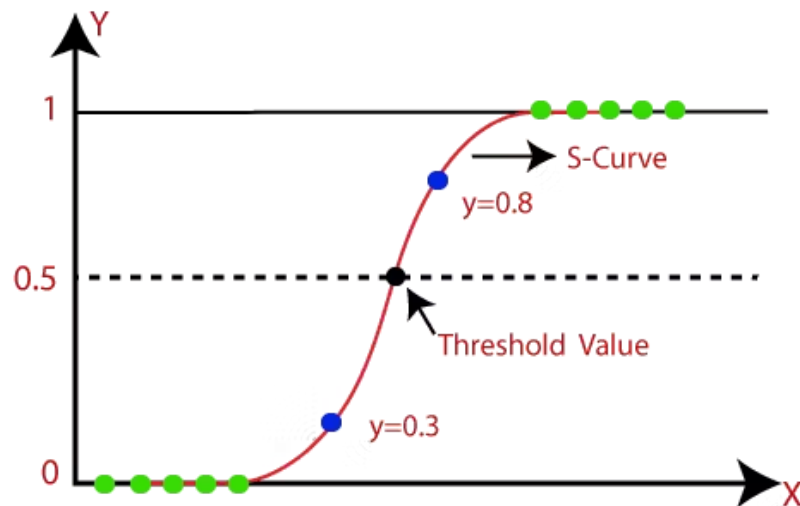


Logistic Regression

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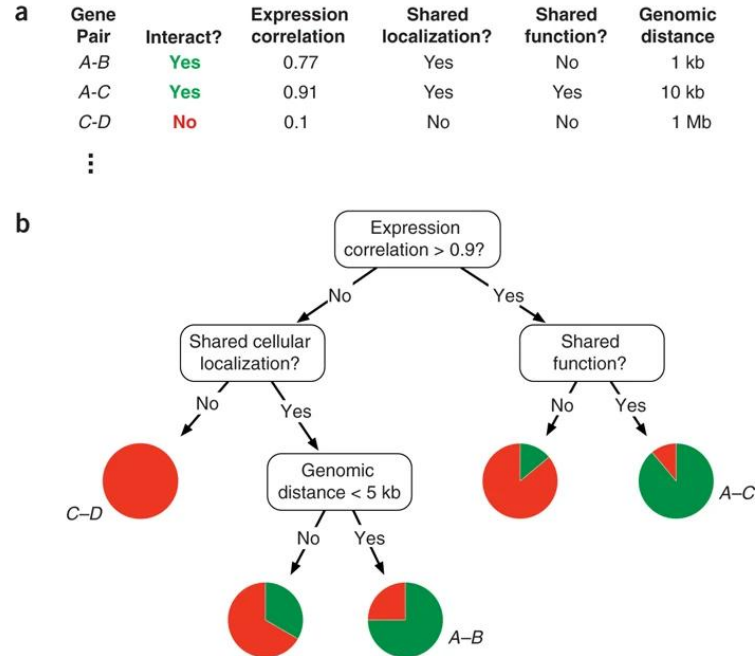
Pros	Cons
• Easy to understand • Good accuracy when data is linearly separable • Less inclined to overfit	• Assumes linearity between output and features • No multicollinearity between features

$$\hat{y}(w, x) = w_0 + w_1x_1 + \dots + w_px_p$$



Decision Trees

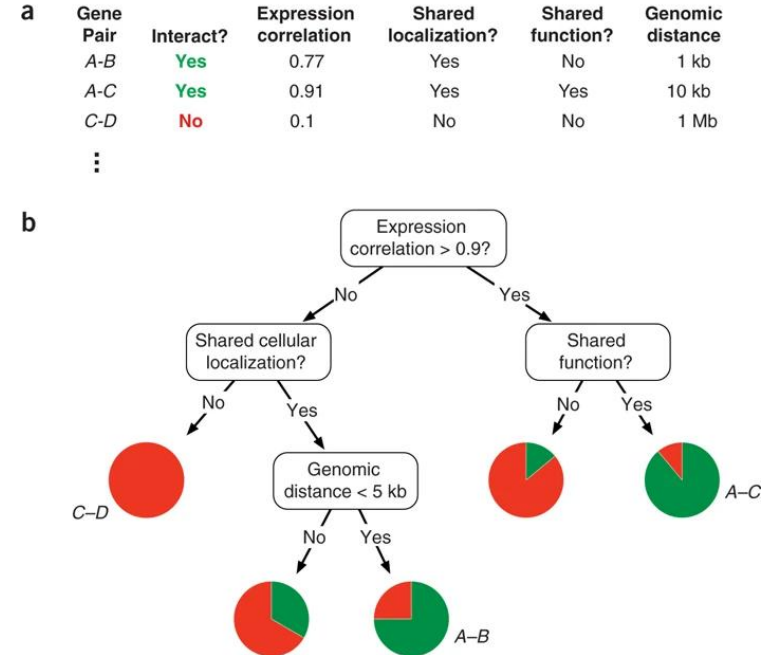
Decision tree learns simple decision rules inferred from features



Decision Trees

Decision tree learns simple decision rules inferred from features

Pros	Cons
• Easy to understand • Can be visualized • Can handle multiple outputs	• Can create overcomplicated trees that overfit • Small variations in data can lead to completely different tree

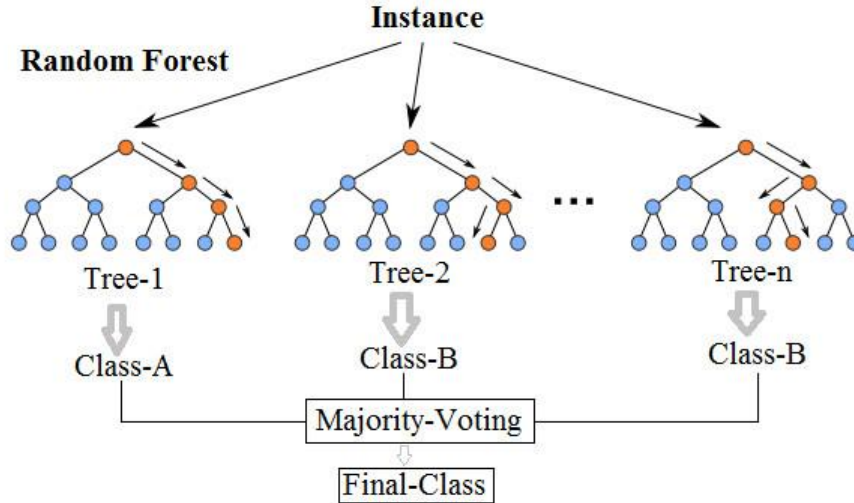


Random Forests

An ensemble of decision trees built from sample of training data

- Prediction is the averaged output of all decision trees

Random Forest Simplified

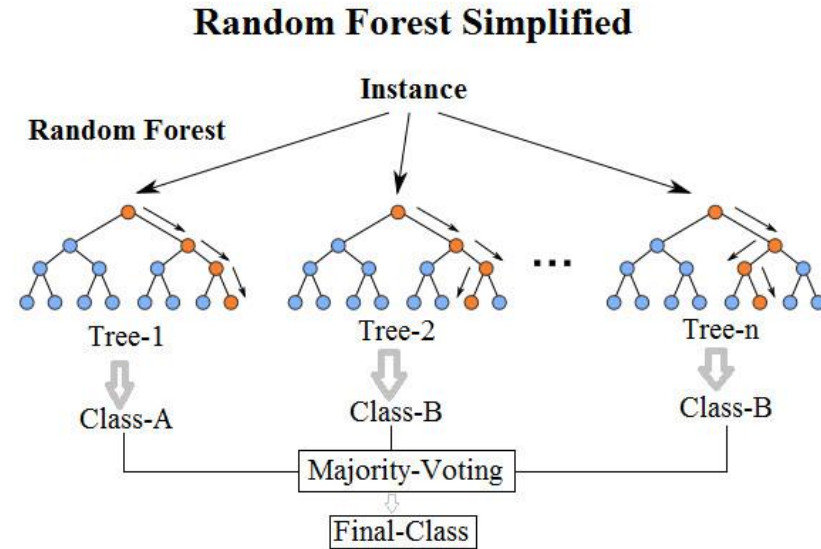


Random Forests

An ensemble of decision trees built from sample of training data

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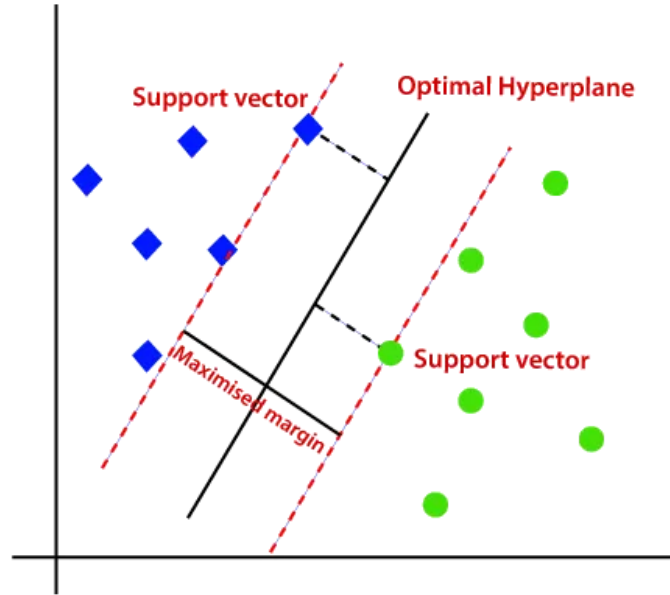
Pros	Cons
● Highly accurate ● Intrinsically interpretable ● Can learn non-linear decision boundaries ● Parallel processing	● High computational complexity ● Can overfit because of noise



Support Vector Machine (SVM)

SVMs find a hyperplane that best separates the data into classes

- Maximize margin between different classes

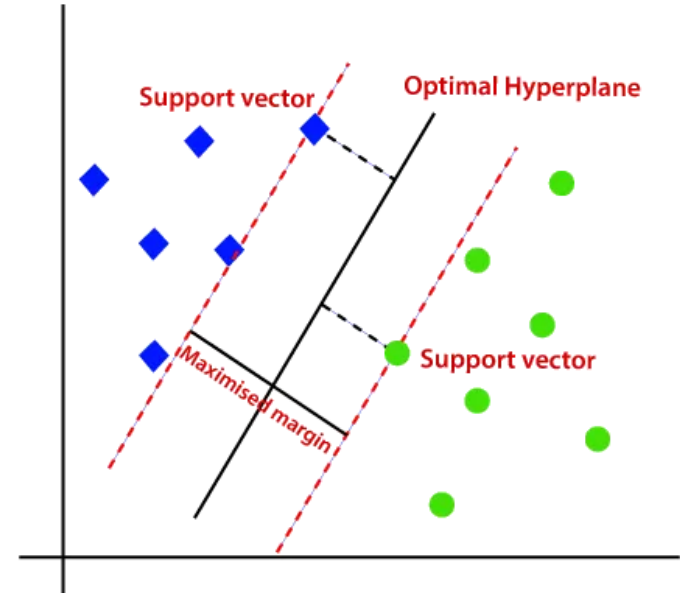


Support Vector Machine (SVM)

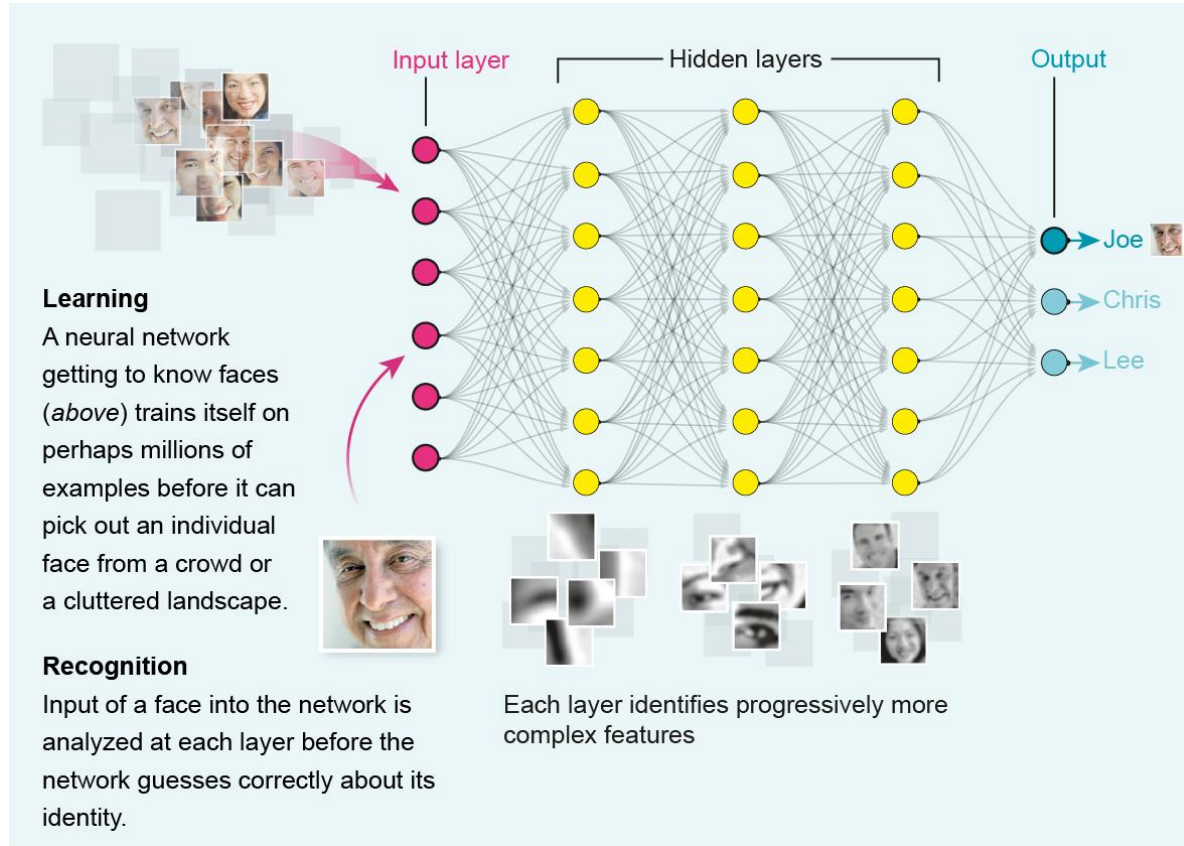
Support Vector Machines find a hyperplane that best separates the data into classes

- Maximize margin between different classes

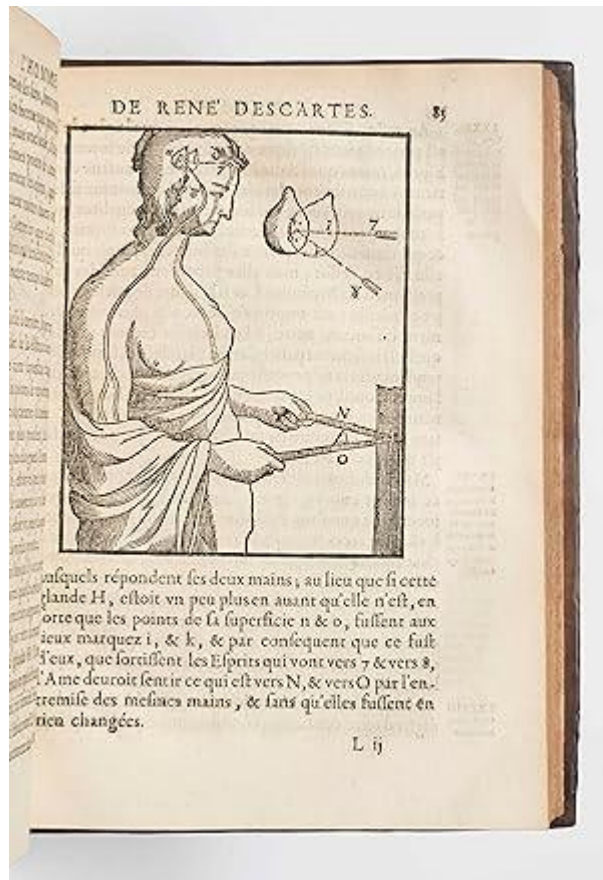
Pros	Cons
• Effectively handles high dimensional data • Can work well with small datasets • Can model non-linear decision boundaries	• Does not work for large data sets • Need to choose kernel • Limited to two-class problems • No probabilistic interpretation



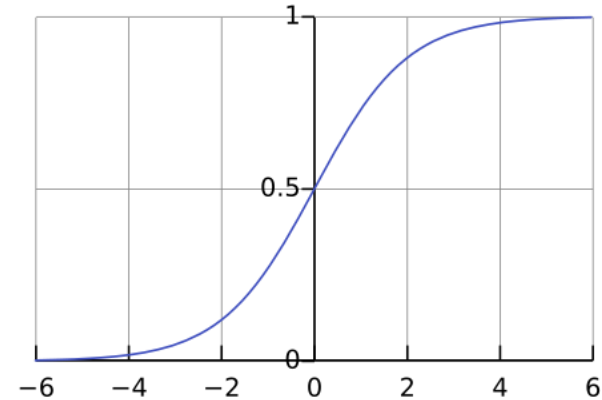
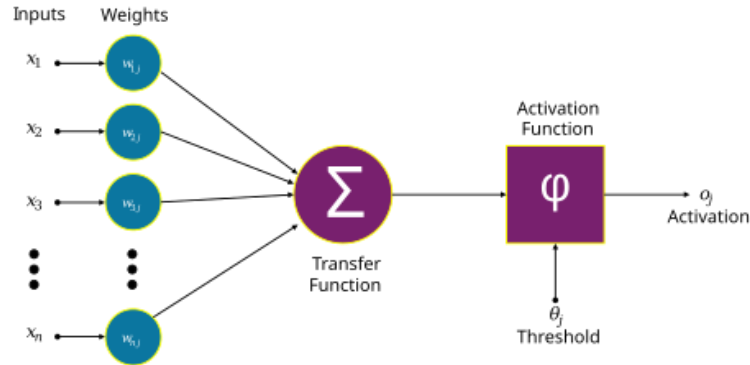
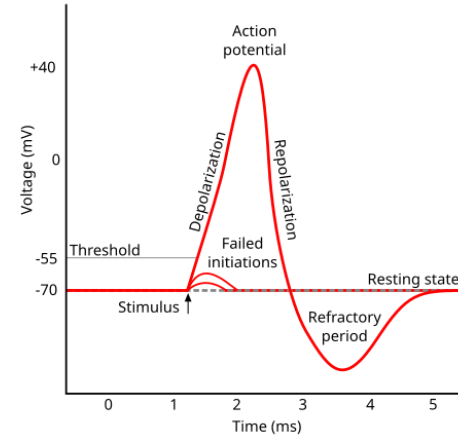
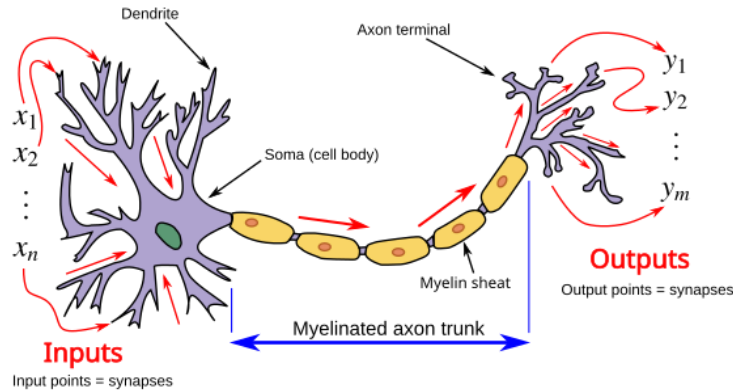
Artificial Neural Networks (ANNs)



Neural Circuitry

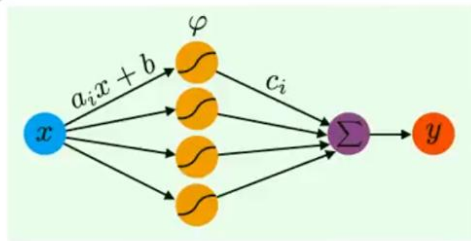


Biological vs Artificial Neurons



The non-linear activation function (sigmoid) is needed to model non-linear relationships

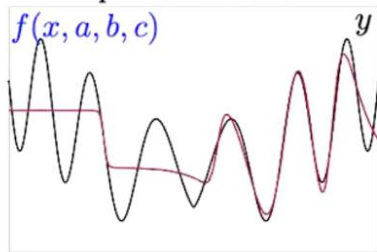
ANNs are “Universal Approximators”



1 hidden layer perceptron:

$$y \approx f(x, a, b, c) \stackrel{\text{def.}}{=} \sum_{i=1}^p c_i \varphi(a_i x + b_i)$$

$p = 6$ neurons



$p = 20$ neurons



Approximation by Superpositions of a Sigmoidal Function*

G. Cybenko†

Abstract. In this paper we demonstrate that finite linear combinations of compositions of a fixed, univariate function and a set of affine functionals can uniformly approximate any continuous function of n real variables with support in the unit hypercube; only mild conditions are imposed on the univariate function. Our results settle an open question about representability in the class of single hidden layer neural networks. In particular, we show that arbitrary decision regions can be arbitrarily well approximated by continuous feedforward neural networks with only a single internal, hidden layer and any continuous sigmoidal nonlinearity. The paper discusses approximation properties of other possible types of nonlinearities that might be implemented by artificial neural networks.

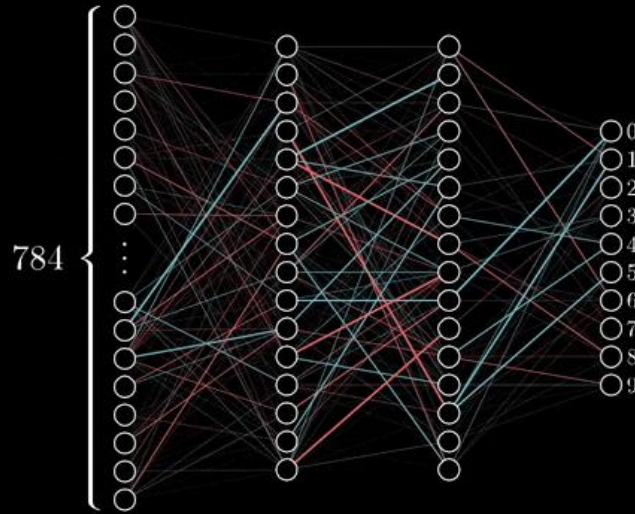
Key words. Neural networks, Approximation, Completeness.



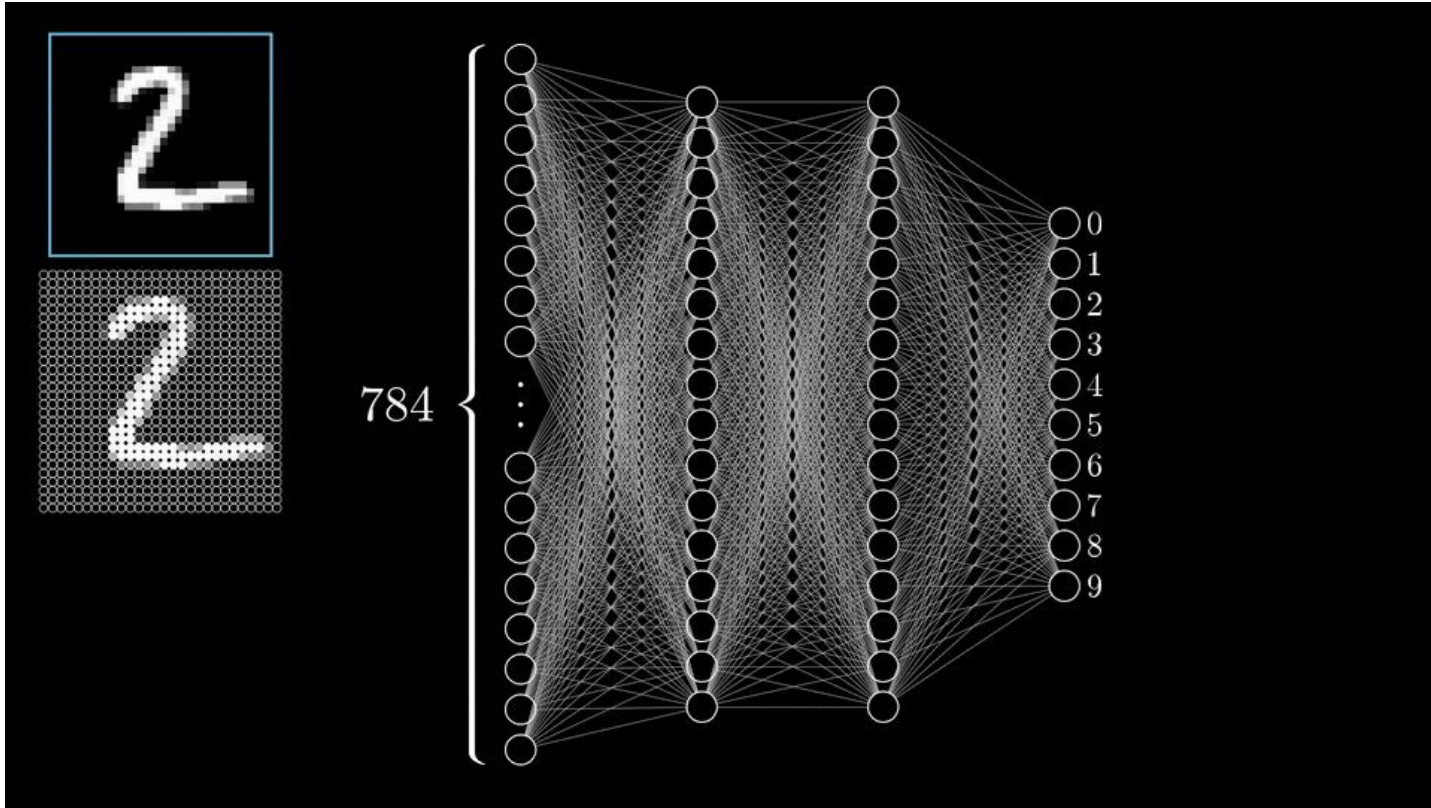
George Cybenko

Training ANNs

Training in
progress...



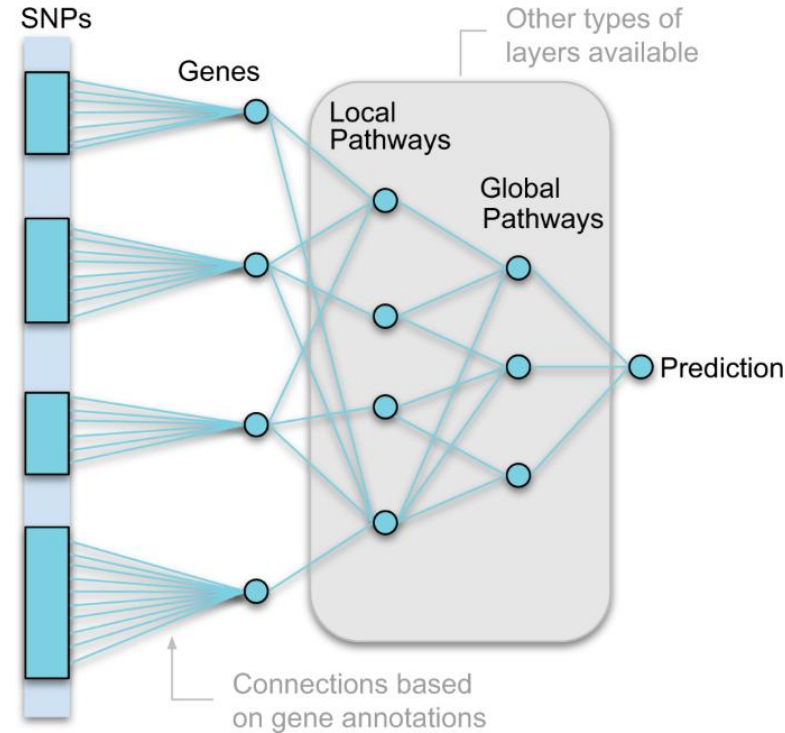
Evaluating ANNs



ANNs in Bioinformatics

GenNet: predicting phenotypes from genetic data

- Connections between layers defined through prior biological knowledge
- Basis for predicting many phenotypes
 - E.g. Schizophrenia, hair and eye color



GenNet framework: interpretable deep learning for predicting phenotypes from genetic data.

van Hilten, A., Kushner, S.A., Kayser, M. *et al. Commun Biol* 4, 1094 (2021).